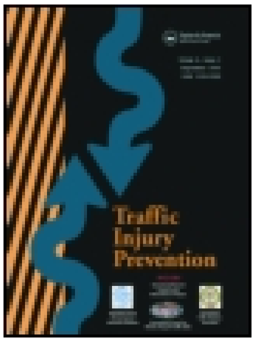

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Safe speed limits for a safe system: The relationship between speed limit and fatal crash rate for different crash types

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SAFE SPEED LIMITS FOR A SAFE SYSTEM: THE RELATIONSHIP BETWEEN SPEED LIMIT AND FATAL CRASH RATE FOR DIFFERENT CRASH TYPES

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ABSTRACT

Objective To provide empirical evidence for ‘safe’ speed limits that will meet the objectives of the Safe System by examining the relationship between speed limit and injury severity for different crash types, using police-reported crash data.

Method Police reported crashes from two Australian jurisdictions were used to calculate a fatal crash rate by speed limit and crash type. Example ‘safe’ speed limits were defined using threshold risk levels.

Results A positive exponential relationship between speed limit and fatality rate was found. For an example fatality rate threshold of 1 in 100 crashes it was found that ‘safe’ speed limits are: 40 km/h for pedestrian crashes, 50 km/h for head on crashes, 60 km/h for hit fixed object crashes, 80 km/h for right angle, right turn and left road / rollover crashes, and 110 km/h or more for rear end crashes.

Conclusions The positive exponential relationship between speed limit and fatal crash rate is consistent with prior research into speed and crash risk. The results indicate that speed zones of 100 km/h or more only meet the objectives of the Safe System, with regard to fatal crashes, where all crash types except rear end crashes are exceedingly rare, such as on a high standard restricted access highway with a safe roadside design.

KEYWORDS

Speed limit, accident rate, accident type, data analysis

INTRODUCTION

The Safe System approach, which is endorsed by the Organisation for Economic Co-operation and Development (OECD) and is the foundation of the *Australian Road Safety Strategy: 2011--2020*, represents the aspiration to create a road transport system in which human mistakes do not result in death or serious injury (Australian Transport Council, 2011). Speed is a key factor affecting the outcome of a human mistake in the road transport system (Elvik *et al.*, 2004). In order to achieve the aspiration of the Safe System, "speeds must be managed so that humans are not exposed to impact forces beyond their physical tolerance." (Australian Transport Council, 2011).

There are several variables that are grouped under the broad category of "speed" in a crash situation. These are: travel speed, impact speed, Delta-V (the change in velocity during the impact) and speed limit. These terms can be thought of as a sequence that contributes to injury severity: speed limit influences travel speed, travel speed in turn influences impact speed, impact speed influences Delta-V, and Delta-V influences injury severity. The speed limit is likely to be the dominant factor that influences travel speed, especially when combined with enforcement that produces general deterrence against exceeding the speed limit. In South Australia the mean travel speeds of vehicles are typically a few km/h below the speed limit, and the 85th percentile speeds typically just above the speed limit (Kloeden and Woolley, 2015). Speed limit changes have also been shown to influence travel speed, though to a lesser extent than the actual numerical change in the speed limit (Musicant, Bar-Gera and Schechtman, 2016). Speed limit could therefore be considered the first step in the sequence towards determination of injury severity in the event of a crash.

Speed limit was chosen as the focus of this analysis for two reasons. First, speed limit is the only speed variable currently set directly by policy, and is therefore the only one that can be directly modified by policy makers. Secondly, data on speed limit and crash severity are readily available to the majority of policy makers in developed countries to perform this type of analysis. By comparison, data on travel speed, impact speed or Delta-V are obtained through detailed crash investigations and crash reconstructions that are generally not conducted on a large scale.

Speed limits, and their relationship to road trauma, have been broadly researched. This has usually been undertaken by examining the effect of a change in speed limit (e.g Farmer, 2016; Hoareau *et al.*, 2006; Mackenzie *et al.*, 2015; Vadeby & Forsman, 2013). The clear majority of these types of studies found that when speed limits were increased, injury crashes also increased, and that when speed limits were decreased, injury crashes also decreased. This prior research has led to the general conclusion that lowering speed limits reduces the severity of crashes. What it has not provided is an answer to the question "what should the speed limit be to achieve the goals of the Safe System approach?"

Research into other speed variables (travel speed, impact speed, Delta-V) has also shown an increase in these speed measures leads to an increase in the number and severity of accidents (Evans, 1994; Augenstein, Perdeck, Stratton, Digges, and Bahouth 2003; Kloeden, McLean, Moore and Ponte 1997; Kloeden, Ponte and McLean 2001; Rosen, Stigson and Sander, 2011; Elvik, Christensen and Amundsen, 2004). Recently, mean travel speeds at crash locations have also been used to examine the concept of 'safe' speed limits for vulnerable road users (Kröyer, 2015a; Kröyer, 2015b).

This paper presents an analysis of police reported crashes from two Australian states to determine if the relationship between speed and injury severity is borne out for speed limits in these data sets. By examining absolute risk, rather than relative risk, the aim is to provide empirical evidence to support calculating the 'safe speed limits' for a given crash type that will meet the objectives of the Safe System philosophy.

METHOD

Crash data were sourced from police reported crashes in two Australian jurisdictions: South Australia (SA) and New South Wales (NSW). Data from 2000 to 2014 for SA and 1996 to 2014 for NSW were used. Crashes on unsealed roads were excluded because it was thought that speed limit may have a much weaker relationship to travel speed, and therefore injury, on such roads. Crashes that involved a heavy vehicle, motorcycle or bicycle were excluded so that the crashes would represent vehicles with broadly similar masses and levels of protection (pedestrian crashes involving car-like vehicles were included for the ‘pedestrian’ crash type). The remaining crash data were disaggregated by crash type. Historically, SA has coded 13 crash types. NSW crash data uses 75 crash types under a coding system known as the Definition for Coding Accidents (DCA). For the purpose of comparison, the NSW data were recoded using DCA codes that matched the crash types used in SA. The SA crash types “side swipe”, “hit parked vehicle”, “hit animal”, “hit object on road”, “other”, and their equivalent NSW DCA codes, were not included in the analysis due to insufficient numbers of fatal crashes. The exclusion criteria reduced the number of crashes included in the analysis from 886,689 to 500,645 (44% excluded) in NSW and from 217,500 to 158,776 (27% excluded) in SA. The number of crashes meeting each exclusion criterion is detailed in Table A1 in the appendix.

The selected SA crash types are shown in Table A2 in the Appendix, along with the corresponding DCA codes used for the NSW data. The crash types “rollover” and “left road – out of control” were combined as it was thought these categories would not be mutually exclusive. No reference to rollover is made in DCA codes, rather a distinction is made between vehicles that depart the road and strike an object, and those that depart the road but do so without striking an object. The DCA codes that relate to road departures where the vehicle did not strike an object were placed in this category, though it is acknowledged this is likely to be an imperfect match.

The denominator used in the fatal crash rate was the number of crashes where a vehicle was towed away or someone was injured, including fatal injuries. This was chosen because the “towed away” variable and injury variable were available in both data sets. One vehicle being towed away is the minimum threshold for reporting a crash to NSW police data. In SA the lower threshold for non-injury cases that are reported to police is a financial value of the property damage, that has varied over time from

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$300 to $3000, or a vehicle being towed. The "towed away" variable was not recorded between 1992 and 1999, and so SA data had to be limited to 2000 onwards.

The numerator used in the fatal crash rate was the number of crashes in which at least one occupant from any vehicle involved in the crash died. In both SA and NSW, the crash is only considered fatal if the person dies as a result of the crash within 30 days of the crash.

The rate of fatal crashes per tow away or injury crashes was then calculated by speed limit for each crash type using Equation 1.

$$Fatal\,crash\,rate\ = \frac{Number\ \ Fatal\,crashes}{Number\ \ tow\,away\,or\,injury\,crashes\left(including\ \ fatal\right)} \quad (1)$$

Results from categories (crash type, speed limit) with fewer than five fatal crashes were excluded to minimise small numbers producing misleading results. The sensitivity of the results to the year the crash occurred was tested by splitting the data from NSW into two time periods. Details of the sensitivity analysis are given in the Appendix.

The calculated fatal crash rates were then used to determine exemplar 'safe' speed limits using example fatal crash rate thresholds (i.e. the rate of fatal crashes per crash being 1 in 100 or 1 in 1000). Given that the data only showed the fatal crash rate at the speed limits currently available, the closest speed limit with a fatal crash rate below the threshold fatal crash rates was defined as the safe speed limit. For cases in which there was disagreement between the SA and NSW data, the safe speed limit was defined as the speed limit for which the average of the SA and NSW fatal crash rates was below the threshold level. If, for any crash type, the fatal crash rates at *any* speed limit available were above the fatal crash rate threshold, the safe speed limit for that crash type was expressed as below the lowest speed limit available. It should be noted that, in practice, a risk threshold would need to be chosen that takes into account the prevalence of different crash types for the road or area in question when determining a 'safe' speed limit.

RESULTS

Log-linear plots of the calculated fatal crash rates by crash type are shown in Figure 1. The fatality rate increases as speed limit increases across all crash types shown, with a few exceptions: hit fixed object crashes and rear end crashes in 110 km/h speed zones relative to 100 km/h zones in NSW; and hit pedestrian crashes in 100 km/h zones relative to 80 km/h zones in SA. The fatality rates of the two states are similar in magnitude and slope for all crash types except “left road out of control / rollover” crashes. The linear appearance of the log-linear plots suggests an exponential relationship between speed limit and fatal crash rate for all the crash types considered. The rates along with the number of crashes by crash type are also shown in Table A3 in the Appendix.

Figure 1 here

Table 1 shows the speed limits that reflect a threshold level of fatal crash rate of either 1 in 100 or 1 in 1000. Insufficient fatal crashes at speed limits low enough to state a 1 in 1000 safe speed limit resulted in these values often needing to be expressed as lower than the lowest speed limit for which data were available. Pedestrian crashes require the lowest speed limit followed by head on crashes, and then hit fixed object crashes. Left road out of control / rollover, right turn and right angle crashes all have similar safe speed limits. Rear end crashes can occur at the highest speed limit in the sample of crashes (110 km/h) while still meeting the 1 in 100 threshold fatal crash rate. It may be possible that this ‘safe’ speed limit is even higher but the lack of data at a speed limit where the fatal crash rate is higher than 1 in 100 results in this remaining unknown. The value is therefore shown in Table 3 as “110 km/h or greater than 110 km/h” as the available data cannot confirm the exact value.

Table 1 here

DISCUSSION

The results have demonstrated a generally positive exponential relationship between speed limit and fatality rate through an analysis of police reported crash data. This positive exponential relationship between speed limit and fatality rate is unsurprising given the reported relationships in prior research on other speed variables, the results of the research into speed limit changes, and the relationship between speed and energy in elementary physics. The similarity of the curves produced by the datasets from two different states provides an additional level of confidence in the results.

As noted in the results, there are a few exceptions to the monotonic relationship between speed limit and fatal crash rate. Two of these occur in 110 km/h zones in NSW and relate to hit fixed object crashes and rear end crashes. The hit fixed object category makes no distinction between the type of object struck, so it seems plausible that the decrease in risk in 110 km/h zone in NSW is due to a greater proportion of less injurious objects, such as barriers, as opposed to trees, being struck. The risk of a rear end crash being fatal is likely to be related to the relative speeds of the vehicle rather than the speed limit, so rear end crashes on 110 km/h roads in NSW may simply be at lower relative speeds due to the restricted level of access permitted on such roads. The difference between NSW and SA in the results for the "left road/rollover" category may also reflect an imperfect match between the SA crash type and the NSW DCA codes. The slight decrease in fatal crash rate for pedestrian crashes in SA from 80 to 100 km/h may be a result of random variation due to the low number of pedestrian crashes in 100 km/h zones (56 crashes, 9 of which were fatal).

The results also show the absolute risk of a fatal crash if a tow away or injury crash occurs. This enables the assessment of speed limits to be made in terms of the goals of the Safe System: defining 'safe' speed limits for which the fatal crash risk is below a threshold that results in virtually zero fatal crashes. Previous work that has attempted to define safe speed limits has used a risk threshold of 1 in 10 when considering impact speed (Wramborg, 2005). Defining the acceptable risk is a difficult task where a variety of approaches can be taken and no single 'correct' answer exists (Hunter & Fewtrell, 2001). The results presented here show safe speed limits using risk thresholds of 1 in 100 and 1 in 1000. These are illustrative only. In practice, a risk threshold would need to be chosen and the prevalence of different crash types for the road or area in question would need to be taken into account when setting the speed limit. Alternatively, there may be a desired speed limit for the road and this can inform the design or modifications so that crash types that would be too dangerous at such a speed limit can be designed out. For example, if head on crashes are occurring on a section of road the speed limit could either be set to 50 km/h, or a median barrier could be installed to prevent this type of crash occurring. In general, if the approach to risk is to follow the Safe System principles, the risk of fatality across the road network should be virtually zero.

Previously safe speeds have been suggested in the form of travel speeds (Tingvall & Haworth, 1999): 30 km/h for pedestrian impacts, 50 km/h for side impacts between cars (right turn, right angle), and 70 km/h for frontal impacts between cars (head on). The results for this present analysis show that head on crashes have a greater risk of fatality than crash types that are likely to lead to side impacts (right angle, right turn), contrary to the suggestion of Tingvall and Haworth. However, the results do agree with Tingvall and Haworth's suggestion that pedestrian crashes require the lowest speed of around 30 km/h. The greater risk associated with head on compared to right angle and right turn crashes can be physically explained by the difference in the change in velocity, Delta-V. Several authors have found that risk of injury is related to Delta-V (Augenstein *et al.*, 2003; Evans, 1994) and that side impacts carry a higher risk of injury than frontal impacts for the same Delta-V (Augenstein *et al.* 2003). Consider then, a vehicle travelling at given speed colliding head on with a vehicle of the same mass, and travelling at the same speed in the opposite direction. Both will experience a Delta-V equal to their travel speed. Alternatively, consider a vehicle that is struck in the side (which is often travelling at low speed) by a vehicle travelling at a given speed. These vehicles will experience a Delta-V of close to half the travel speed of the striking vehicle (given simplifying assumptions such as a fully plastic impact and equal masses). In addition, it is sometimes assumed that head on crashes are always frontal crashes but head on crashes can involve a side impact if a vehicle has lost control prior to the impact with the other vehicle. The greater closing speed in head on crashes may also result in less time for the driver to react and reduce the vehicle speed prior to impact. All these factors potentially contribute to head on crashes having a greater risk of being fatal, at a certain speed limit, than right turn or right angle crashes.

Many jurisdictions, including Australia, have speed limits of 100 km/h or more. The analysis suggests that speed limits of 100 km/h or more would only meet the objectives of the Safe System where all crash types except rear end crashes are exceedingly rare, such as on a high standard divided road with grade separated junctions and a safe roadside design. Collisions with barriers may also be tolerable in high-speed zones but this analysis did not specifically identify the type of object struck and therefore does not inform on this topic. This analysis has considered risk of fatality when a crash occurs: it does not consider the risk of a crash occurring. Reducing the risk of crashes occurring through road infrastructure improvements or semi-autonomous vehicle technologies would allow the use of higher speed limits while still achieving a Safe System. In effect, this would allow the use of a lower threshold level of fatal crash rate to be used.

The absolute risk of fatal crash relative to speed limit is likely to be influenced by compliance with the speed limit. Therefore the validity of generalising the results to other jurisdictions will depend upon the relative levels of speed limit compliance. Speed limit compliance may vary due to factors such as perceived reasonableness of speed limits, and speed limit enforcement practices. In South Australia compliance with speed limits is high; generally more than 90% of drivers travel less than 5 km/h above the speed limit (Kloeden and Woolley, 2015). NSW reports very high compliance past speed camera sights (greater than 99%), but has not published details of general compliance (Centre for Road Safety, 2017). In areas with lower levels of speed limit compliance the absolute risk at a given speed limit will be increased. For areas that differ substantially it would be best to replicate the analysis with crash data from the area in question.

Police reported crash data from two Australian states show a general positive exponential

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relationship between speed limit and fatal crash rate, consistent with prior research on other speed variables and crash risk. Absolute risk values were calculated that provide an empirical basis for Safe System speed limits. The possibility of pedestrian crashes justifies the lowest speed limit, followed by head on crashes, hit fixed object crashes, right turn and right angle crashes, and rear end crashes. The analysis does not inform on the safety of impacts with barriers. It suggests that speed zones of 100 km/h or more only meet the objectives of the Safe System, with regard to fatal crashes, where all crash types except rear end crashes are exceedingly rare, such as on a high standard restricted access highway with a safe roadside design.

Limitations

A limitation of the analysis is the large span of time from which the data are taken, dating back to 2000 in SA and 1996 in NSW. Data were drawn from a large time span to obtain sufficient numbers of fatal crashes; however, factors that affect the fatality rate independent of speed limit and crash type have likely changed over this time. For example, it has been shown that the crashworthiness of vehicles is improving over time (Anderson & Doecke, 2010; Newstead *et al.*, 2013). Changes in the distribution of speed limits over the time span of the data may also affect the results. If certain speed limits are more predominant in recent years this may cause a reduction in fatality rate, compared to other speed limits, which is not a function of the speed limit but of the decreasing fatality rate. This would certainly be the case for 50 km/h zones, which were only introduced in 2002 in SA and 1997 in NSW. The use of 110 km/h zones has also been reduced in SA in recent years with speed limits being reduced to 100 km/h on 1,100 km of road in 2003 and a further 723 km in 2011 (Dua, Anderson, Cartwright and Holmes 2013). The sensitivity analysis assessing the effect of year of crash on fatality risk (see Appendix) revealed that the general relationship between speed limit and fatal crash risk remained consistent, but that the absolute risk values can change over time, which can mean an increase in what is determined to be the 'safe' speed limit.

A further limitation of the analysis is that it does not control for type or standard of road. While the main influence of type and standard of road is likely to be on crash frequency and the relative proportion of the different crash types that occur, neither of which influence the analysis performed, they may have some influence of the severity of certain crash types. The severities of single vehicle crashes (rollover / left road out of control, hit fixed object) are most likely to be influenced by the type or standard of the road, but it may also have some effect on other crash types.

A limitation of the data used in the analysis was lack of a clearly defined serious injury category of injury severity. To examine the stated goals of the Safe System approach, it would be best to produce a measure of serious and fatal injury rate. However, this is difficult to measure from police reported data where serious injury is often poorly defined or recorded. It is likely that this is a limitation of many such datasets. Precise injury coding, such as the abbreviated injury score (AIS), is often only used for much smaller datasets due to the time consuming nature of such coding and the detailed injury information required. The data from SA include the injury categories: treated by doctor, treated at hospital, admitted to hospital, and fatal. The NSW historically only contains the injury categories "injury" and "fatal". While it might appear that SA data contain a measure of serious injury, "admitted to hospital", previous examinations of the data have shown that misclassification between "admitted to hospital" (serious injury) and "treated at hospital" (minor injury) is not uncommon. Consequently, only the fatal crash rate was examined.

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Table 1: The safe speeds limits for a given threshold fatal crash rate.

Crash type	1 in 100 safe speed limit (km/h)	1 in 1000 safe speed limit (km/h)
Hit pedestrian	40	<40
Head on	50	<50
Hit fixed object	60	<50
Right angle	80	50
Right turn	80	50
Left road / rollover	80	<60
Rear end	110 or >110	80

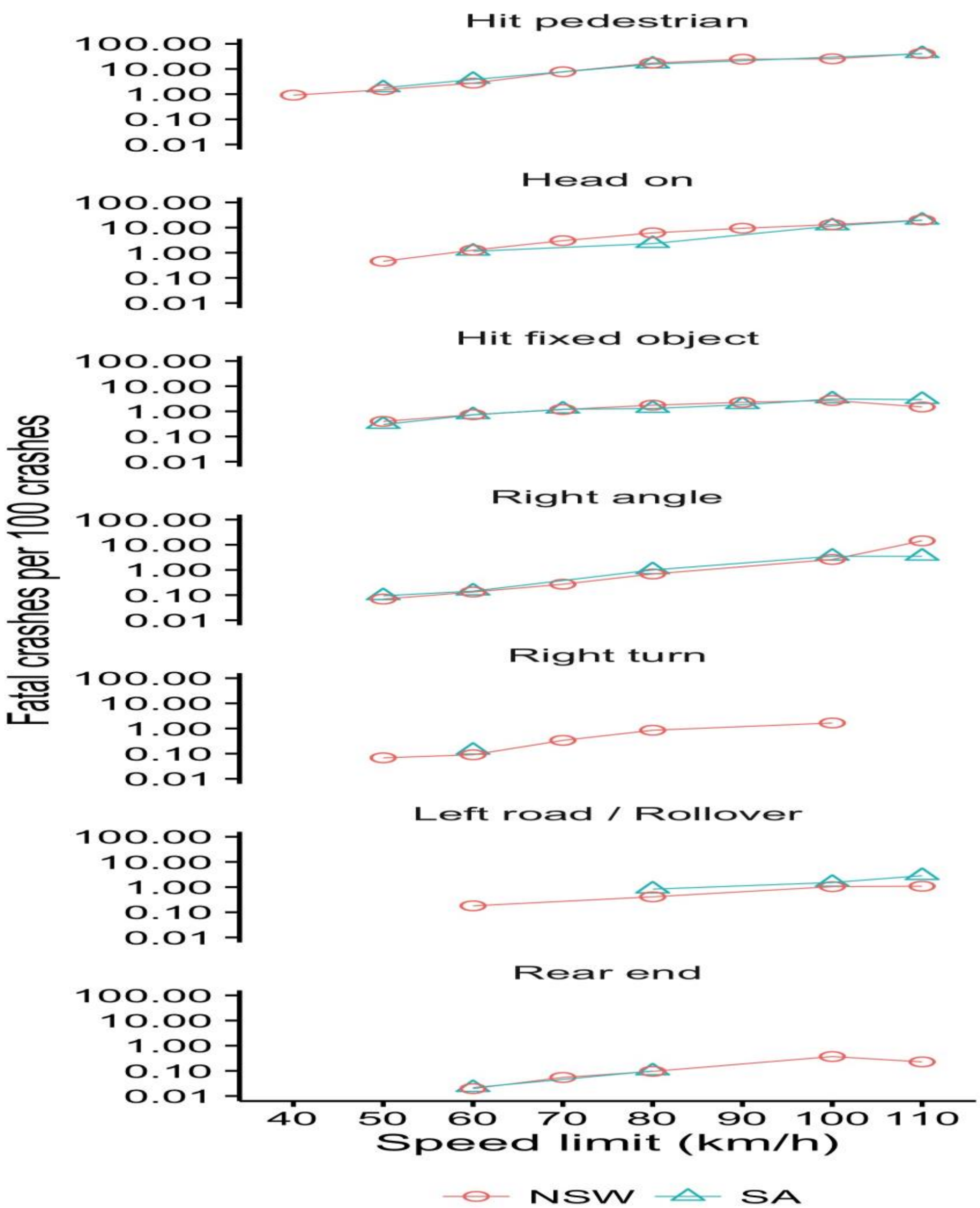

Figure 1: The fatal crash rates by speed limit for different crash types in NSW and SA

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